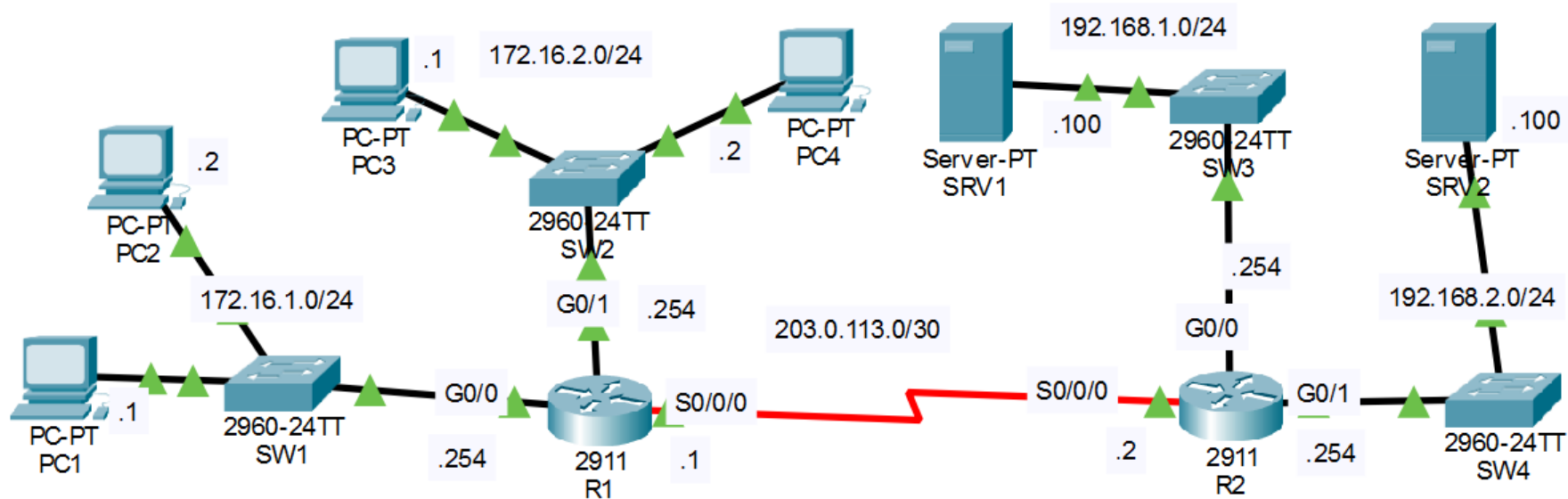




1. Configure OSPF on R1 and R2 to allow full connectivity between the PCs and servers.
2. Configure standard numbered ACLs on R1 and standard named ACLs on R2 to fulfill the following network policies:
 - Only PC1 and PC3 can access 192.168.1.0/24
 - Hosts in 172.16.2.0/24 can't access 192.168.2.0/24
 - 172.16.1.0/24 can't access 172.16.2.0/24
 - 172.16.2.0/24 can't access 172.16.1.0/24

Network Topology + Tasks

If you require further assistance please contact us by sending email to export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R1>en

R1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#router ospf 1

R1(config-router)#net 172.16.0.0 0.0.255.255 area 0

R1(config-router)#net 203.0.113.0 0.0.0.255 area 0

R1(config-router)#do sh ip ospf int

GigabitEthernet0/0 is up, line protocol is up

Internet address is 172.16.1.254/24, Area 0

Process ID 1, Router ID 203.0.113.1, Network Type BROADCAST, Cost: 1

Transmit Delay is 1 sec, State DR, Priority 1

Designated Router (ID) 203.0.113.1, Interface address 172.16.1.254

No backup designated router on this network

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Hello due in 00:00:08

Index 1/1, flood queue length 0

Next 0x0(0)/0x0(0)

Last flood scan length is 1, maximum is 1

Last flood scan time is 0 msec, maximum is 0 msec

Neighbor Count is 0, Adjacent neighbor count is 0

Suppress hello for 0 neighbor(s)

GigabitEthernet0/1 is up, line protocol is up

Internet address is 172.16.2.254/24, Area 0

Process ID 1, Router ID 203.0.113.1, Network Type BROADCAST, Cost: 1

Transmit Delay is 1 sec, State DR, Priority 1

Designated Router (ID) 203.0.113.1, Interface address 172.16.2.254

No backup designated router on this network

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Hello due in 00:00:08

Index 2/2, flood queue length 0

Next 0x0(0)/0x0(0)

Last flood scan length is 1, maximum is 1

Last flood scan time is 0 msec, maximum is 0 msec

Neighbor Count is 0, Adjacent neighbor count is 0

Suppress hello for 0 neighbor(s)

Serial0/0/0 is up, line protocol is up

Internet address is 203.0.113.1/30, Area 0

Process ID 1, Router ID 203.0.113.1, Network Type POINT-TO-POINT, Cost: 64

Transmit Delay is 1 sec, State POINT-TO-POINT,

Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

Configured OSPF on R1 and verified
OSPF is active on both LAN interfaces

DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up

Configure OSPF on R2 and establish a
neighbor relationship with R1.

R2>

R2>en

R2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router ospf 1

R2(config-router)#network 192.168.0.0 0.0.255.255 area 0

R2(config-router)#network 203.0.113.0 0.0.0.255 area 0

R2(config-router)#

10:20:54: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.1 on Serial0/0/0
from LOADING to FULL, Loading Done

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Verified the OSPF neighbor relationship between R2 and R1, showing a FULL adjacency.

```
R2(config-router)#do sh ip ospf neigh
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
203.0.113.1	0	FULL/ -	00:00:31	203.0.113.1	Serial0/0/0

```
R2(config-router)#
```



R1



```
10:20:54: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.2 on Serial0/0/0 from LOADING
to FULL, Loading Done
```

Verified the OSPF routing table on R1, showing
OSPF-learned routes to both 192.168.1.0/24 and
192.168.2.0/24.

```
R1(config-router)#
```

```
R1(config-router)#do sh ip ro
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
```

```
C      172.16.1.0/24 is directly connected, GigabitEthernet0/0
```

```
L      172.16.1.254/32 is directly connected, GigabitEthernet0/0
```

```
C      172.16.2.0/24 is directly connected, GigabitEthernet0/1
```

```
L      172.16.2.254/32 is directly connected, GigabitEthernet0/1
```

```
O      192.168.1.0/24 [110/65] via 203.0.113.2, 00:02:54, Serial0/0/0
```

```
O      192.168.2.0/24 [110/65] via 203.0.113.2, 00:02:54, Serial0/0/0
```

```
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C      203.0.113.0/30 is directly connected, Serial0/0/0
```

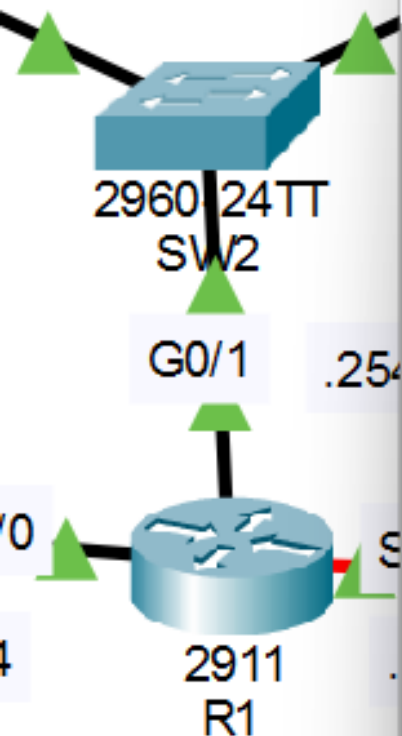
```
L      203.0.113.1/32 is directly connected, Serial0/0/0
```

```
R1(config-router)#
```

[Copy](#)[Paste](#)[Top](#)

Verified the OSPF routing table on R2, showing
OSPF-learned routes to both 172.16.1.0/24 and
172.16.2.0/24.

172.16.2.0/24



to allow full connect

CLS on R1 and sta
policies:

s 192.168.1.0/24

ccess 192.168.2.0

172.16.2.0/24

R2

Neighbor ID	Pri	State	Dead Time	Address	Interface
203.0.113.1	0	FULL/ -	00:00:31	203.0.113.1	Serial0/0/0

R2(config-router)#do sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/24 is subnetted, 2 subnets

O	172.16.1.0/24	[110/65]	via 203.0.113.1, 00:01:38, Serial0/0/0
O	172.16.2.0/24	[110/65]	via 203.0.113.1, 00:01:38, Serial0/0/0

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C	192.168.1.0/24	is directly connected, GigabitEthernet0/0
L	192.168.1.254/32	is directly connected, GigabitEthernet0/0

192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C	192.168.2.0/24	is directly connected, GigabitEthernet0/1
L	192.168.2.254/32	is directly connected, GigabitEthernet0/1

203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks

C	203.0.113.0/30	is directly connected, Serial0/0/0
L	203.0.113.2/32	is directly connected, Serial0/0/0

--More--

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```
172.16.0.0/24 is subnetted, 2 subnets
O   172.16.1.0/24 [110/65] via 203.0.113.1, 00:01:38, Serial0/0/0
O   172.16.2.0/24 [110/65] via 203.0.113.1, 00:01:38, Serial0/0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.1.0/24 is directly connected, GigabitEthernet0/1
L   192.168.1.254/32 is directly connected, GigabitEthernet0/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.2.0/24 is directly connected, GigabitEthernet0/1
L   192.168.2.254/32 is directly connected, GigabitEthernet0/1
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C   203.0.113.0/30 is directly connected, Serial0/0/0
L   203.0.113.2/32 is directly connected, Serial0/0/0
```

```
R2(config-router)#
R2(config-router)#
R2(config-router)#
R2(config-router)#
R2(config-router)#
R2(config-router)# ip access-list standard TO_192.168.1.0/24
R2(config-std-nacl)#permit 172.16.1.1
R2(config-std-nacl)#permit 172.16.2.1
R2(config-std-nacl)#deny any
R2(config-std-nacl)#int g0/0
R2(config-if)#ip access-group TO_192.168.1.0/24 out
R2(config-if)#
```

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Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Request timed out.

Reply from 192.168.1.100: bytes=32 time=6ms TTL=126

Reply from 192.168.1.100: bytes=32 time=6ms TTL=126

Reply from 192.168.1.100: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.1.100:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>

Verified that PC1 can access SRV1 (192.168.1.100) through the configured ACL.



PC2



Physical

Config

Desktop

Programming

Attributes

Command Prompt



Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 203.0.113.2: Destination host unreachable.

Reply from 203.0.113.2: Destination host unreachable.

Reply from 203.0.113.2: Destination host unreachable.

Reply from 203.0.113.2: Destination host unreachable.

Ping statistics for 192.168.1.100:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|

Verified that PC2 is blocked from accessing SRV1
(192.168.1.100) by the ACL.

Configured an ACL on R2 to deny hosts in 172.16.2.0/24 from accessing the 192.168.2.0/24 network.

```
R2 (config-if) #
R2 (config-if) #
R2 (config-if) # ip access-list standard TO_192.168.2.0/24
R2 (config-std-nacl) # deny 172.16.2.0 0.0.0.255
R2 (config-std-nacl) # permit any
R2 (config-std-nacl) # int g0/1
R2 (config-if) # ip access-group TO_192.168.2.0/24 out
R2 (config-if) #
```

61528 PAKETS 110001 TO COMMAND LINE 110

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Request timed out.

Reply from 192.168.1.100: bytes=32 time=6ms TTL=126

Reply from 192.168.1.100: bytes=32 time=6ms TTL=126

Reply from 192.168.1.100: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.1.100:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>ping 192.168.2.100

Pinging 192.168.2.100 with 32 bytes of data:

Request timed out.

Reply from 192.168.2.100: bytes=32 time=6ms TTL=126

Reply from 192.168.2.100: bytes=32 time=6ms TTL=126

Reply from 192.168.2.100: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.2.100:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>|

Verified that PC1 can access both SRV1 (192.168.1.100) and SRV2 (192.168.2.100), confirming permitted connectivity.



Send packet 12345 to command line 110

C:\>ping 192.168.2.0

Pinging 192.168.2.0 with 32 bytes of data:

Reply from 203.0.113.2: bytes=32 time=1ms TTL=254

Reply from 203.0.113.2: bytes=32 time=6ms TTL=254

Reply from 203.0.113.2: bytes=32 time=7ms TTL=254

Reply from 203.0.113.2: bytes=32 time=6ms TTL=254

Ping statistics for 192.168.2.0:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 7ms, Average = 5ms

C:\>ping 192.168.2.100

Verified PC3 is blocked from accessing SRV2.

Pinging 192.168.2.100 with 32 bytes of data:

Reply from 203.0.113.2: Destination host unreachable.

Reply from 203.0.113.2: Destination host unreachable.

Reply from 203.0.113.2: Destination host unreachable.

Reply from 203.0.113.2: Destination host unreachable.

Ping statistics for 192.168.2.100:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|



“Configured standard numbered ACLs on R1 to restrict traffic between the 172.16.1.0/24 and 172.16.2.0/24 networks.

```
R1>
R1>
R1>
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#access-list 1 deny 172.16.1.0 0.0.0.255
R1(config)#access-list 1 permit any
R1(config)#
R1(config)#
R1(config)#access-list 2 deny 172.16.2.0 0.0.0.255
R1(config)#access-list 2 permit any
R1(config)#
```

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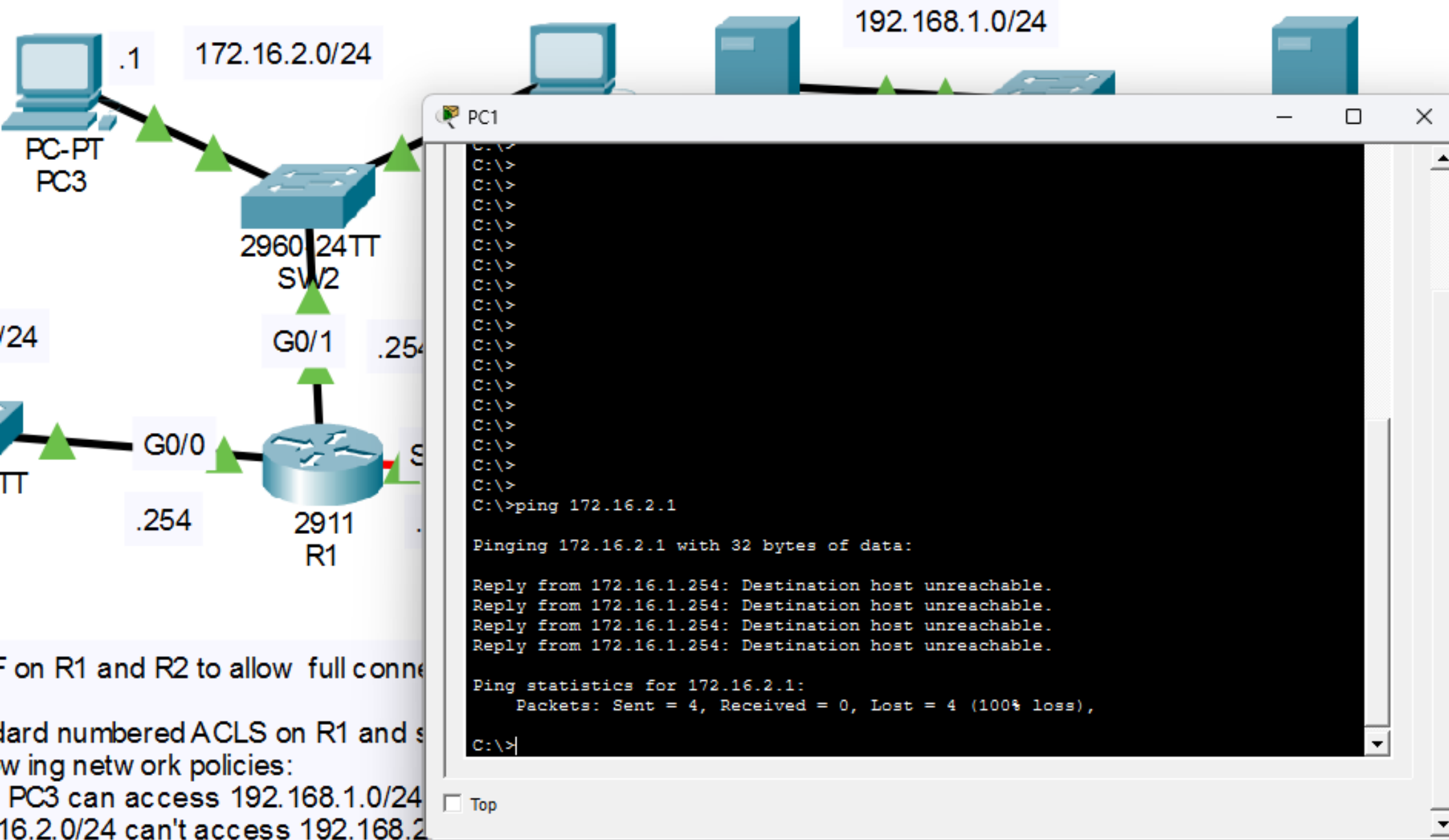
Verified the standard ACLs on R1, confirming both network restrictions are configured

```
R1 (config) #  
R1 (config) #do sh access-list  
Standard IP access list 1  
  10 deny 172.16.1.0 0.0.0.255  
  20 permit any  
Standard IP access list 2  
  10 deny 172.16.2.0 0.0.0.255  
  20 permit any  
R1 (config) #
```

Applied the standard ACLs to R1 interfaces to enforce traffic restrictions.

```
R1 (config) #  
R1 (config) #  
R1 (config) #int g0/1  
R1 (config-if) #ip access-group 1 out  
R1 (config-if) #  
R1 (config-if) #  
R1 (config-if) #int g0/0  
R1 (config-if) #ip access-group 2 ou  
R1 (config-if) #ip access-group 2 out  
R1 (config-if) #
```

Verified that PC3 is blocked from accessing the 172.16.1.0/24 network





PC4



Command Prompt



Cisco Packet Tracer PC Command Line 1.0

C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.2.254: Destination host unreachable.

Reply from 172.16.2.254: Destination host unreachable.

Reply from 172.16.2.254: Destination host unreachable.

Reply from 172.16.2.254: Destination host unreachable.

Ping statistics for 172.16.1.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

Verified that PC4 is blocked from accessing PC2.